

EXPERIENCE

IIoT Software Engineer

January 2024 — Present

Schlumberger (SLB)

Houston, TX

- Engineered the edge platform's production **authentication stack**, unblocking a **\$50M line of business** that could not ship without it, by building a **Keycloak** and **OIDC** identity provider that authenticates users on edge devices whether or not the device is online, a **Go** credential-management sidecar, and the TLS certificate service securing service-to-service traffic across the fleet.
- Expanded** the platform's qualified hardware by **roughly 3x** and **enabled a new revenue stream** valued at **over \$100M**, by executing **end-to-end testing and integration** of new edge hardware and **diagnosing** driver, firmware, and protocol interoperability for **5G modems** and **Wireless modules defects** alongside engineers at four partner vendors.
- Accelerated** the edge telemetry reporting path **500x** while eliminating read-write contention, by overhauling it into an asynchronous pipeline with runtime-configurable redundant queues and replacing lock-contended shared state with concurrency-safe structures.
- Spearheaded a **security research program** end to end as **project lead**, from landscape survey through stakeholder recommendation, investigating which **autonomous AI agents** find real vulnerabilities without human direction and whether they can absorb the first pass of a paid external penetration test, by evaluating six agentic frameworks through a shared model gateway against a purpose-built application spanning the full OWASP Top 10, scored on a rubric locked before any run.
- Systematized data exchange between industrial field devices and cloud systems, introducing a single standardized path for equipment speaking **MQTT, OPC UA, Modbus TCP and RTU, and NATS**, by building containerized edge services in **Go, Python, C#, and C++** deployed across **thousands** of edge devices for oil and gas clients.
- Introduced authenticated **real-time** access to the edge message bus for external and browser-based clients, by engineering a **WebSocket-to-NATS** proxy implementing **OIDC token verification** with cached JWKS key resolution, connection keepalive, and buffer and backpressure management.

IIoT Software Engineering Intern

June 2023 — August 2023

Schlumberger (SLB)

Houston, TX

- Built the shared cloud-messaging path for every application on the industrial gateway, removing the need for each one to implement and maintain its own cloud transport, by engineering a **containerized Python service** that bridges the gateway's internal message bus to a TLS-secured cloud **MQTT** broker, supporting all bus data types with automatic reconnection on connection loss.
- Eliminated static network configuration from gateway deployment, by implementing **Zeroconf/mdNS service** discovery across two codebases — the Python connector and the C# broker it pairs with — so the two services locate each other automatically on the gateway network.
- Launched the connector to production in **seven weeks** as its sole developer, by building the service, a data generator for development testing, separate production and debug container images with on-device SSH troubleshooting, the edge application manifest, and the build pipeline.

Software Engineering Intern

June 2022 — December 2022

Zededa Inc.

Remote

- Built the container deployment **reference for developers** in ZEDED's Developer Program, giving them a working example of containerized workloads running on distributed edge devices, by packaging a web application as a Docker image deployable to EVE-OS and documenting the build and deployment procedure through Zedcontrol.
- Assembled a virtual machine appliance that ran a complete third-party application stack on edge hardware with no manual installation, by provisioning an Ubuntu virtual machine for an open-source school administration application, configuring its networking, and exporting it to the QCOW2 format EVE-OS deploys.
- Clarified EVE setup, hardware onboarding, and application lifecycle management for customers, enabling those workflows to be completed in Zedcontrol without escalating to support, by preparing product and API documentation published in the ZEDED Help Center.

EDUCATION

Bachelor of Science in Computer Science, University of Maryland, Baltimore County (UMBC)

2023

- Minor: Economics
- Honor: Cum Laude
- GPA: 3.675 / 4.0

SKILLS

Languages	Go, Python, C/C++, C#, x86 & ARM Assembly, HTML, CSS, R, Git, LaTeX
ML Libraries	Numpy, Scikit-learn, TensorFlow, PyTorch, Pandas, Matplotlib
Hardware	RPi, Industrial PCs, Servers, VMs, VDE
Operating System	Windows, MacOS, Linux, Unix, EVE